Securing understanding of arithmetic sequences

1 The arithmetic (linear) sequence 17, 21, 25, 29, 33, ... has a(n) __________ of +4. Tick **1** correct answer

- [ ] n^{th} term
- [ ] 1^{st} term
- [x] common difference
- [ ] multiplier

2 Which of these sequences are arithmetic? Tick **3** correct answers

- [x] 2, 6, 10, 14, ...
- [x] 2, 7, 12, 17, ...
- [ ] 2, 8, 15, 23, ...
- [ ] 2, 4, 8, 16, ...
- [x] 2, -1, -4, -7, ...

3 Match the n^{th} term to the first five terms of each sequence. Write the correct letter in each box

a	$3n + 2$
b	$2n + 3$
c	$6n - 1$
d	$6 - n$
e	$5 + n$
f	$1 - 6n$

d	$5, 4, 3, 2, 1, \dots$
b	$5, 7, 9, 11, 13, \dots$
a	$5, 8, 11, 14, 17, \dots$
f	$-5, -11, -17, -23, -29, \dots$
e	$6, 7, 8, 9, 10, \dots$
c	$5, 11, 17, 23, 29, \dots$

4 The solution to the two-step equation $8x - 13 = 83$ is $x =$ 12 Fill in the blank

5 Which is true of the solution to the equation $3x - 10 = 3$ Tick **1** correct answer

- [ ] It is a positive integer.
- [ ] It is a negative integer.
- [x] It is non-integer.
- [ ] It has no solution.

6 $8, 13, 18, 23, 28, \ldots$ are the first five terms of which sequence? Tick **1** correct answer

- [ ] $8n + 5$
- [ ] $5n + 8$
- [x] $5n + 3$
- [ ] $5n - 3$